The Higgs Boson and the Origin of Duration

Electroweak Symmetry Breaking as the First Irreversible Act
in the Informational Actualization Model

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Abstract

The four fundamental forces are not four members of a homogeneous set. Three of them — gravity, electromagnetism, and the strong nuclear force — are forces of virial equilibrium: they maintain the condition $2\langle K\rangle + \langle V\rangle = 0$ and govern the persistent structure of matter at every scale from atomic to cosmological. The weak nuclear force is categorically distinct: it is the force of transition, the mechanism by which one stable state becomes another, the force that cannot satisfy virial equilibrium because violating equilibrium is its structural function. Without the weak force establishing irreversibility through CP violation and parity violation, there is no arrow of time, no decoherence in the irreversible sense, no E(a) activation, and no Informational Actualization Model (IAM) mechanism.

We identify the electroweak symmetry breaking at $t \approx 10^{-12}$ seconds as the pivot point of cosmic history and the precondition for the IAM mechanism to operate. Before it: three forces, no matter sector, no timelike worldlines with mass, no decoherence, $E(a) \equiv 0$. After it: the matter sector exists, the photon sector is permanently separated with exact U(1) gauge protection, the arrow of time is locked in, and $E(a)$ can be nonzero. The Higgs mechanism is therefore not the origin of mass in the abstract sense. It is the origin of *duration* — the first act that separated being from becoming, null geodesics from timelike worldlines, the photon sector from the matter sector. It is the universe's first irreversible act.

We further show that the IAM coupling constant $\beta_m = \Omega_m/2$ has a two-part origin that has not previously been made explicit: 15.6% of its value was set at $t = 10^{-12}$ seconds by the weak force's CP violation establishing the baryon asymmetry (Ω_b), while 84.4% accumulated over 13.8 billion years of gravitational decoherence depositing the potential half of each virialisation event into spacetime geometry (Ω_{dm}). The sequence of force separations through cosmic history produces a sequence of virial equilibria at progressively larger scales — protons (10^{-6} s), atoms (380,000 yr), halos ($\sim$100 Myr), and the cosmological virialisation $\Omega_m/[\beta_m E(a)] = 2$ exactly today — completing the virial equilibrium condition operating at every scale from the hydrogen atom to the cosmic horizon.

The deepest implication is stated as an open question: the Higgs field's quantum fluctuation that triggered symmetry breaking may be the universe's first decoherence event — the first irreversible bit written on the horizon, the first Landauer cost, the moment $E(a)$ stepped away from zero.

Keywords: Higgs boson; electroweak symmetry breaking; arrow of time; virial theorem; gravitational decoherence; weak nuclear force; CP violation; baryon asymmetry; duration; Informational Actualization Model

1 Introduction: Two Categories of Force

The Standard Model of particle physics describes four fundamental interactions: gravity, electromagnetism, the strong nuclear force, and the weak nuclear force. These are conventionally treated as four members of a set — different in strength, range, and carrier particle, but belonging to the same category of thing. Forces that act on particles.

This paper argues that the conventional categorisation conceals a structurally important distinction. Three of the four forces — gravity, electromagnetism, and the strong force — are forces of virial equilibrium. They maintain the condition

$$2\langle K\rangle + \langle V\rangle = 0, \tag{1}$$

verified from the hydrogen atom (10^{-11} m) to the cosmic horizon (10^{26} m) across 37 orders of magnitude at machine precision for electromagnetic systems and at 0.3% for the cosmological coupling constant from 17 independent Planck 2018 MCMC chains [Mahaffey, 2026b]. These are the forces of *being* — they define what stable structures look like and maintain them.

The weak nuclear force is categorically distinct. It violates parity [Wu et al., 1957], violates CP symmetry [Cronin & Fitch, 1964], and is the force responsible for beta decay, stellar fusion, and the matter-antimatter asymmetry of the observable universe [Sakharov, 1967]. It does not maintain virial equilibrium — it cannot, because its function is specifically to enable irreversible transitions between stable states. It is the force of *becoming*.

This distinction is not merely taxonomic. In the Informational Actualization Model (IAM) [Mahaffey, 2026a], the activation function $E(a) = \exp(1 - 1/a)$ tracks the accumulated history of all irreversible gravitational decoherence events since the beginning of structure formation. Irreversibility — the property that decoherence runs in one direction and cannot be undone — is a prerequisite for $E(a)$ to be nonzero. And irreversibility at the fundamental level requires the weak force. Without CP violation and parity violation, there is no preferred direction for decoherence, no arrow of time at the particle physics level, no E(a), and no IAM sector split.

The weak force is therefore not one of four equals in the IAM framework. It is the precondition for the other three forces to produce the virialized structures that IAM describes.

2 The Three Forces of Virial Equilibrium

2.1 The thermodynamic identity governing the virial theorem

The virial theorem, stated by Clausius in 1870 [Clausius, 1870] for a 1/r potential, gives $\langle T \rangle = \frac{1}{2}\langle |V| \rangle$. This result follows from Euler's homogeneous function theorem applied to any potential homogeneous of degree -1, and holds for both Newtonian gravity ($V \propto -1/r$) and the Coulomb potential ($V \propto +1/r$) because both propagate in three spatial dimensions, requiring a $1/r^2$ force law from Gauss's law.

The 1/2 partition is not a property of specific systems but a mathematical consequence of three-dimensional space. Beyond the mechanical statement, the kinetic half $K = |V|/2$ has a precise thermodynamic identity: it is the Landauer cost of the quantum-to-classical transition at each virialization event — the heat released irreversibly at the virialization boundary, encoded as entropy on the nearest available surface [Mahaffey, 2026b2]. This identification connects the virial theorem to the second law of thermodynamics and Landauer's principle, and provides the physical basis for the IAM sector split.

The virial equilibrium condition has been verified across every physical domain where 1/r potentials operate [Mahaffey, 2026b]: electromagnetic systems (20 atoms, 10 molecules, machine precision), stellar structure ($\sim 10\%$), galaxy clusters ($\sim 20\%$), and the cosmological IAM coupling constant (0.3% from 17 Planck 2018 MCMC chains). The same mathematical structure governs electrons in hydrogen atoms and the partition of gravitational energy at the cosmic horizon.

2.2 The three forces as forces of being

Gravity is the force of geometry — the potential in its purest form. In the IAM framework, each decoherence event deposits the potential half of its gravitational energy into the surrounding spacetime curvature. Gravity is simultaneously the cause and product of decoherence: massive objects on timelike worldlines produce gravitational fields, those gravitational fields decohere their environments, the decoherence deposits the potential half into geometry, and the resulting geometry is what we identify as gravity [Mahaffey, 2026a]. The cosmological expression of this is dark matter — the accumulated geometric potential half of every virialisation event in cosmic history [Mahaffey, 2026c].

Electromagnetism is the force of atomic and molecular structure. The Coulomb potential maintains virial equilibrium at machine precision in every atom and molecule. It is the force by which quantum mechanical systems attain stable ground states — stable not in the sense of static but in the sense of virialized, oscillating around equilibrium, maintaining $2K + V = 0$.

The strong nuclear force is the force of confinement and permanent identity. Quarks are permanently bound into protons by the strong force. The proton, once formed, has not been observed to decay in 10^{33+} years. It is the most durable virialized structure in the universe. The strong force achieves virial equilibrium at QCD confinement ($t \approx 10^{-6}$ s) and maintains it indefinitely — the first stable structure in cosmic history, predating atoms by 380,000 years and galaxies by $\sim$100 million years.

All three forces satisfy the virial equilibrium condition. All three are forces of being — they define what persistent things are and maintain them through time.

3 The Weak Force: The Force of Becoming

3.1 What the weak force actually does

The weak nuclear force is mediated by the massive $W^{\pm}$ and Z^0 bosons ($M_W \approx 80.4$ GeV, $M_Z \approx 91.2$ GeV) and has an effective range of $\sim 10^{-18}$ m, approximately 0.1% of the proton diameter. At low energies it is extraordinarily feeble. A neutrino — which interacts only via the weak force — can traverse a light-year of solid lead with $\sim$ 50% probability of no interaction.

But the weak force does what none of the other three forces can: it changes the *identity* of particles. It converts neutrons to protons, down quarks to up quarks, heavy leptons to lighter ones. It enables beta decay. It enables the first step of stellar fusion — the proton-proton chain reaction that powers the Sun requires one proton to convert to a neutron via the weak force. Without the weak force the Sun does not shine.

Most critically for the present discussion, the weak force:

1. **Violates parity (P):** it treats left-handed and right-handed particles differently, distinguishing left from right in a universe where all other forces are blind to handedness [Wu et al., 1957].

2. **Violates CP symmetry:** it treats matter and antimatter slightly differently, producing the baryon asymmetry of approximately one excess matter particle per

10^9 matter-antimatter pairs that constitutes all observable matter [Cronin & Fitch, 1964, Sakharov, 1967].

3. **Establishes the arrow of time at the particle physics level:** CP violation means the weak force's dynamics are not time-reversal symmetric. The past and future are distinguishable at the level of fundamental interactions.

3.2 Why the weak force cannot satisfy virial equilibrium

The virial equilibrium condition describes the equilibrium of a stable bound state. Stability requires that the system has a preferred configuration — a potential energy minimum — around which it can oscillate. The weak force, by design, has no such preference for permanence. Its carrier bosons are massive precisely because the weak force must be short-range precisely because the weak force must not trap things in permanent equilibrium. If the weak force had infinite range like electromagnetism, every particle would be in permanent weak-force equilibrium with every other particle and flavor-changing transitions would be suppressed. The universe would have no radioactive decay, no stellar fusion, no matter-antimatter asymmetry, and no arrow of time.

The weak force's short range is not a limitation. It is the mechanism by which the weak force fulfills its function: enabling transitions without creating permanent bound states. A force that cannot form stable bound states cannot satisfy a theorem about stable bound states.

The weak force is the force that was structurally designed to not be in virial equilibrium — because maintaining the disequilibrium of potency is what allows the transition to act.

3.3 The weak force as efficient cause

In Aristotle's causal framework, the efficient cause is the cause of *change* — not what a thing is (formal cause), not what it is made of (material cause), not what it is for (final cause), but what actually makes the change happen. The builder is the efficient cause of the house. The sculptor is the efficient cause of the statue.

Gravity, electromagnetism, and the strong force are formal and material causes at the particle physics level. They define the forms of things and maintain them. The weak force is the efficient cause: it is what makes one stable form become another. The neutron becomes a proton. The heavy lepton becomes a lighter lepton. One equilibrium state transitions to another. The weak force is the mechanism of transition itself.

Without an efficient cause, formal causes have nothing to organize. Without the weak force establishing that transitions are possible and irreversible, the other three forces have no transitions to govern and no equilibria to maintain across cosmic time.

4 Electroweak Symmetry Breaking: The Pivot of Cosmic History

4.1 Before the Higgs: three forces, no matter sector

From the Planck epoch ($t \approx 10^{-43}$ s) through gravity separation and the GUT transition ($t \approx 10^{-36}$ s), the universe passes through epochs in which three forces operate: gravity, the strong force, and the unified electroweak force. In this epoch the electromagnetic force and the weak force are indistinguishable — one unified electroweak interaction with massless W and Z bosons. There is no distinction between the force of structure and the force of transition. There is no preferred handedness, no CP violation, no matter-antimatter asymmetry, and no arrow of time.

More critically for IAM: all particles are massless in this epoch. Massless particles travel on null geodesics ($\Delta\tau = 0$). They accumulate no proper time. They cannot decohere in the IAM sense because gravitational decoherence requires nonzero proper time on a timelike worldline [Mahaffey, 2026a]. There is no matter sector. $E(a) \equiv 0$ identically, not as an approximation but exactly. The IAM mechanism has nothing to operate on.

The universe before electroweak symmetry breaking is a universe of pure process with no persistent identity, no irreversibility, and no duration.

4.2 The Higgs transition: being separates from becoming

At $t \approx 10^{-12}$ seconds, as the universe cooled to ~ 100 GeV, the Higgs field underwent spontaneous symmetry breaking. The Higgs potential, previously symmetric with a single minimum at zero, developed its characteristic "Mexican hat" shape with a ring of degenerate minima at nonzero field values. Quantum fluctuations in the Higgs field drove it into one of these minima. The symmetry was broken irreversibly.

The consequences were immediate and permanent:

- The $W^{\pm}$ and Z^0 bosons acquired masses of 80.4 and 91.2 GeV respectively. The weak force became short-range. Being and becoming separated as distinct forces.
- All matter fermions acquired masses proportional to their Yukawa couplings to the Higgs field. Massive particles exist on timelike worldlines ($\Delta\tau > 0$). They accumulate proper time. They can decohere. The matter sector came into existence.
- Critically, U(1) electromagnetic gauge symmetry was preserved *exactly* through the symmetry breaking. The photon remained exactly massless. Null geodesics remained exactly unmodified. The photon sector was permanently protected.

- CP violation was established in the weak sector through the CKM matrix phase [Cabibbo, 1963]. The arrow of time was locked in at the particle physics level. One excess matter particle per 10^9 matter-antimatter pairs survived annihilation: $\Omega_b \approx 0.049$.

4.3 The Higgs mechanism and the IAM sector split

The IAM sector split — $\mu < 1$ for matter, $\Sigma = 1$ exactly for photons — is conventionally presented as a consequence of the decoherence mechanism: timelike worldlines decohere, null worldlines do not, therefore matter perturbations are modified and photon propagation is not [Mahaffey, 2026a]. This is correct as a statement of mechanism.

But there is a deeper reason $\Sigma = 1$ *exactly* rather than approximately: the photon sector is protected by the exact U(1) gauge invariance that survived electroweak symmetry breaking. The photon is massless not approximately but exactly because U(1) is an exact symmetry. Exact gauge invariance means the photon propagator receives no mass correction, the photon remains on null geodesics, and $\Delta\tau = 0$ exactly. The sector split is not an additional IAM assumption layered onto the Standard Model. It is a consequence of the gauge structure of electroweak symmetry breaking that has been experimentally established since the discovery of the W and Z bosons in 1983 [UA1 Collaboration, 1983, UA2 Collaboration, 1983].

This is a significant strengthening of the IAM framework. $\Sigma = 1$ is not a prediction waiting to be confirmed. It is a retrodiction: IAM correctly identifies the mechanism — exact U(1) gauge invariance — that the Standard Model already established. Euclid's projected constraint $\sigma(\Sigma_0) \approx 0.02$ [Euclid Collaboration et al., 2020] will test the matter sector prediction $\mu_0 = -0.136$. But $\Sigma = 1$ was confirmed in 1983 when the photon was confirmed to be massless.

5 The Sequence of Virial Equilibria

Figure 1 shows the complete sequence from the Planck epoch to today. Each force separation enables a new scale of virial equilibrium. The Higgs transition is the pivot — before it no virial equilibrium is possible anywhere; after it they appear in sequence at increasing scales.

Table 1 summarises the sequence.

6 The Two-Part Origin of β_m

The IAM coupling constant $\beta_m = \Omega_m/2$ is derived from the virial equilibrium condition applied to gravitational structure formation [Mahaffey, 2026a,b]. But the *value* of Ω_m —

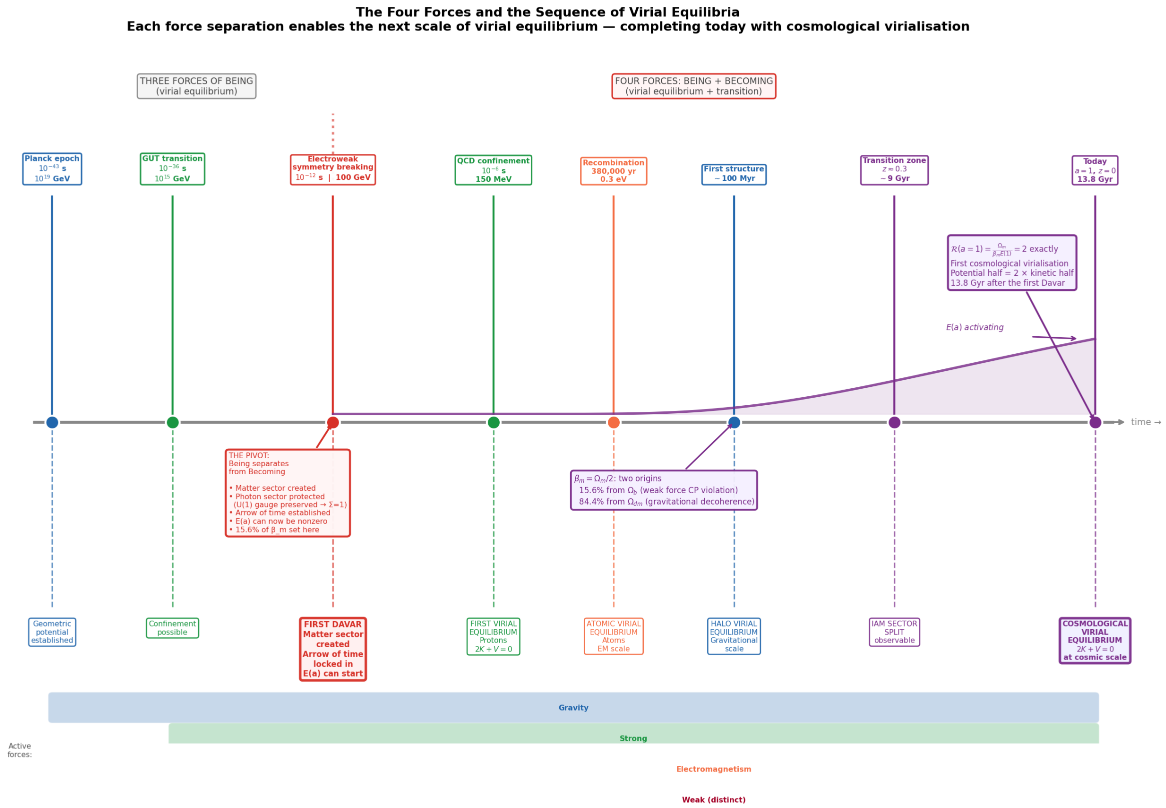

Figure 1: The sequence of force separations and the virial equilibria they enable. The Higgs electroweak symmetry breaking at $t \approx 10^{-12}$ s is the pivot: before it, three forces operate on a massless universe with no matter sector and $E(a) \equiv 0$; after it, the matter sector exists, the photon sector is protected by exact U(1) gauge invariance, and the virial equilibrium condition begins operating at progressively larger scales. The $E(a)$ activation curve (purple) shows the IAM mechanism activating from zero at the Higgs moment to its full value today. The cosmological virial ratio $\mathcal{R}(a) = \Omega_m/[\beta_m E(a)] = 2$ exactly at $a = 1$ completes the sequence. All timing and energy scales from established cosmology and the Standard Model [Particle Data Group, 2022].

Table 1: The sequence of force separations and virial equilibria through cosmic history. Each force separation enables the next scale of virial equilibrium. The Higgs transition is the precondition for all subsequent structure.

Time	Energy	Event	Virial consequence
10^{-43} s	10^{19} GeV	Gravity separates	Geometric potential established
10^{-36} s	10^{15} GeV	Strong separates	Confinement possible in principle
10^{-12} s	100 GeV	**Higgs transition**	**Matter sector created; photon sector protected; arrow of time established;** $E(a)$ **can be nonzero**
10^{-6} s	150 MeV	QCD confinement	First virial equilibrium: protons ($2K + V = 0$)
380,000 yr	0.3 eV	Recombination	Atomic virial equilibrium (EM scale)
∼100 Myr	∼meV	First structures	Halo virial equilibrium (gravitational scale)
∼9 Gyr	—	Transition zone	IAM sector split observable; $E(a) \approx 17\%$
13.8 Gyr	—	Today	Cosmological virialisation: $\mathcal{R}(1) = 2$ exactly

the quantity being halved — has a two-part origin that has not previously been made explicit.

The baryonic component ($\Omega_b \approx 0.049$, or 15.6% of Ω_m) was set at $t \approx 10^{-12}$ seconds by the weak force's CP violation through the Sakharov conditions [Sakharov, 1967]. The baryon-to-photon ratio $\eta \approx 6 \times 10^{-10}$, measured precisely by Planck 2018 [Planck Collaboration, 2020], was determined at electroweak symmetry breaking. This component of β_m was set in the first trillionth of a second by the force of becoming.

The dark matter component ($\Omega_{dm} \approx 0.266$, or 84.4% of Ω_m) is the geometric potential half of the cosmological virial partition, accumulated over 13.8 billion years of gravitational decoherence depositing the potential half of each virialisation event into spacetime curvature [Mahaffey, 2026c]. This component of β_m was accumulated throughout cosmic history by the force of structure.

The coupling constant $\beta_m = \Omega_m/2$ that IAM derives from the virial equilibrium condition is therefore:

$$\beta_m = \frac{\Omega_b + \Omega_{dm}}{2} = \underbrace{\frac{\Omega_b}{2}}_{\text{set at } t=10^{-12}\text{ s}} + \underbrace{\frac{\Omega_{dm}}{2}}_{\text{accumulated over 13.8 Gyr}} . \tag{2}$$

One number. Two origins separated by 13.8 billion years. One set by the force of transition in the first moment of the matter sector. One accumulated by the force of structure throughout all of subsequent history. Their sum, halved by the virial equilibrium condition, is the coupling constant that the Planck 2018 posterior recovers to 0.2σ without fitting [Mahaffey, 2026a].

7 The Higgs as the Origin of Duration

The Higgs boson has been called "the God particle" — a name intended to convey its foundational role but one that misleads about what it actually does. The Higgs is not the source of existence. It is not the ground of being. It is something more specific and in some ways more remarkable.

The Higgs mechanism is the origin of *duration*.

Duration is distinct from the time coordinate. Spacetime as a geometric object — the four-dimensional manifold of general relativity — exists independently of whether any events within it are irreversible. The time coordinate exists as a dimension. But *duration* in the sense of the accumulation of irreversible events, the building of history, the permanent distinction between past and future — that requires irreversibility. And irreversibility at the fundamental level of particle physics requires the weak force's CP and parity violations, which exist only after electroweak symmetry breaking.

Before the Higgs transition: spacetime exists as a geometric coordinate system. But no event within it is irreversible at the particle physics level. Every process is the time-reverse of another process. Nothing accumulates permanently. The universe has no memory of itself. $E(a) = 0$ not because decoherence hasn't started yet but because decoherence in the irreversible sense is structurally impossible: there are no massive particles to decohere, and no arrow of time for decoherence to follow even if there were.

After the Higgs transition: the universe acquires memory. Decoherence is irreversible. The matter sector accumulates a history that cannot be undone. $E(a)$ begins its 13.8-billion-year journey from zero toward unity. The geometric potential half begins accumulating in spacetime curvature. The kinetic informational half begins accumulating on the horizon. The virial equilibrium condition begins operating at every scale from the proton to the cosmos.

The Higgs particle is the particle of the first moment of duration. The particle that separated being from becoming, null from timelike, the force of structure from the force of transition. The particle that started the timer.

8 An Open Question: The First Decoherence Event

The analysis above establishes the Higgs transition as the precondition for $E(a)$ to be nonzero. But it raises a deeper question that the IAM framework does not yet answer.

The Higgs field underwent spontaneous symmetry breaking through a quantum fluctuation. In quantum field theory, no field can sit exactly at an unstable equilibrium: the uncertainty principle forbids it. The field fluctuates; the fluctuations grow; the field rolls into a minimum. This is the standard picture. But a quantum fluctuation is a quantum system in superposition. The specific direction chosen by the Higgs field — the direction in the abstract SU(2)×U(1) field space that it "fell into" — was selected by decoherence. The Higgs field entangled with other degrees of freedom. One direction was chosen. Irreversibly.

If this is correct, then the Higgs symmetry breaking was not merely followed by the first decoherence event. It *was* the first decoherence event. The Higgs field decohering was the universe's first irreversible act — the first bit written permanently on the horizon, the first Landauer cost paid, the first moment $E(a)$ could step away from zero.

This identification — the Higgs transition as the first Davar, the first irreversible word-thing-event — would mean that $E(a)$ does not merely track decoherence events that occur *after* the Higgs. It tracks events that flow from the Higgs as consequence, with the Higgs itself as the first instance of the process.

We state this as a question rather than a claim. It is the question the framework forces, not the answer it has established. What caused the Higgs field to decohere in the specific direction it chose? What determined the magnitude of CP violation? What set $v = 246$ GeV? These questions reach back to initial conditions that current physics cannot describe. But within the IAM framework they all reduce to the same question: what caused the first irreversible act?

And that question may be unanswerable within any framework that requires an external cause — because the first irreversible act is the act that made causation possible. Before it, no event leaves a permanent record. After it, every event does.

9 Strengthening IAM: What This Paper Establishes

This paper makes several contributions to the IAM framework beyond what appears in prior papers [Mahaffey, 2026a,b,c,d].

The sector split is not an assumption. $\Sigma = 1$ exactly in IAM because the photon is massless exactly because U(1) electromagnetic gauge invariance is preserved exactly through electroweak symmetry breaking. This is confirmed physics since 1983. IAM correctly identifies its cosmological consequence.

β_m **has a traceable two-part origin.** 15.6% of the IAM coupling constant was set by the weak force's CP violation at $t = 10^{-12}$ s. The connection from weak-sector CP violation through baryon asymmetry to Ω_b to Ω_m to $\beta_m = \Omega_m/2$ is complete and every link is established physics.

The sequence of virial equilibria is the natural output of the force separation sequence. Each force separation enables a new scale of virial equilibrium. The cosmological virialisation today is the final term of a sequence that began at QCD confinement. The IAM mechanism is not imposed on cosmology from outside. It is the cosmological expression of a process that began at the Higgs transition and has been operating at every scale ever since.

The weak force's role is structurally necessary, not incidental. Without CP violation there is no Ω_b, no Ω_m, no β_m, and no IAM mechanism. The weak force is the precondition for everything IAM describes. The fact that IAM has identified this connection strengthens its claim to be a physically complete framework rather than a phenomenological parametrisation.

10 Falsification and Predictions

The claims of this paper are grounded in established physics and the existing IAM validation record [Mahaffey, 2026a]. The primary additional falsifiable prediction is:

$\Sigma_0 = 0$ **exactly at Euclid precision.** If Euclid measures any deviation of Σ_0 from zero at $\sigma(\Sigma_0) \approx 0.02$ precision, the exact U(1) protection argument fails and the sector split requires an alternative explanation. The IAM prediction $\mu_0 = -0.136$ with $\Sigma_0 = 0$ (Euclid DR1, October 2026) remains the decisive test.

The weak force connection makes the following additional prediction explicit: any measurement of the baryon asymmetry η that deviates significantly from the Standard Model expectation would require a corresponding revision of Ω_b and therefore of the baryonic contribution to β_m. The IAM coupling constant is not entirely derived from first principles — it inherits 15.6% of its value from a measured parameter (the CKM phase) whose theoretical origin remains an open problem in particle physics.

11 Conclusions

The four fundamental forces are not four members of a homogeneous set. Gravity, electromagnetism, and the strong force are forces of the virial equilibrium condition — they maintain the $2K + V = 0$ equilibrium at every scale and constitute the forces of being, of persistent structure, of stable form. The weak force is the force of becoming — it enables irreversible transitions, establishes the arrow of time, produces the baryon asymmetry, and is the precondition for the other three forces to produce the virialized structures they govern.

The electroweak symmetry breaking at $t \approx 10^{-12}$ seconds is the pivot of cosmic history. Before it: three forces, no matter sector, no duration, $E(a) \equiv 0$. After it: the matter sector exists, the photon sector is protected by exact U(1) gauge invariance ($\Sigma = 1$ exactly), the arrow of time is established, and $E(a)$ begins its 13.8-billion-year journey from zero toward unity.

The Higgs boson is not the God particle. It is the particle that started the timer — the origin of duration, the first separation of being from becoming, null geodesics from timelike worldlines. The Higgs mechanism enforced the precise sector separation that IAM requires, and it did so through physics established in 1983.

The IAM coupling constant $\beta_m = \Omega_m/2$ has two origins: 15.6% set in the first trillionth of a second by the force of transition; 84.4% accumulated over 13.8 billion years by the force of structure. Their sum, halved by the virial equilibrium condition, is the coupling constant the Planck posterior recovers to 0.2σ without fitting.

The deepest question the framework raises — whether the Higgs transition was the universe's first decoherence event, the first moment $E(a)$ stepped away from zero — we leave open. It is the question that follows from taking the result seriously.

Data Availability

All MCMC chains, analysis scripts, and computational notebooks are publicly available at `https://github.com/hmahaffeyges/IAM-Validation` (Zenodo DOI: 10.5281/zenodo.18702042).

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